CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

CHARUSAT Campus, Changa – 388 421.

M Sc (BIOCHEMISTRY)
SEMESTER II EXAMINATION (REGULAR)
BC705: IMMUNOLOGY

Date: 03.05.2011
Time: 10.00 AM to 1.00 PM **Total Marks: 70**

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A **Total marks: 20**

1. Conversion of pre-TCR to TCR induces the formation of double positive immature thymocyte. **(True / False)**
2. A precipitate of antigen and antibody can be formed with monovalent Fab fragment. **(True / False)**
3. Clip is bound to both MHC I and MHC II during the process of antigen presentation. **(True / False)**
4. Radial immunodiffusion is a quantitative method. **(True / False)**
5. CD3 complex is associated with BCR for activation and signal transduction of B cell. **(True / False)**
6. Lymph node is responsible for activation and differentiation of both T and B cells. **(True / False)**
7. Exocytosis of CTL granule from cytotoxic T cell releases ------------------ and granzyme at T cell-target cell junction.
8. Immunotoxin is synthesized by replacing ----------------subunit with specific monoclonal antibody
   a) Receptor binding subunit b) toxic subunit c) both subunits d) none of them
9. Constant region determinants in an antibody, that collectively define each heavy-chain class and subclass are called
   a) Isotype b) allotype c) idiotype d) none of these
10. The immunoglobulin class that binds with very high affinity to membrane receptors on basophils and mast cells is: a) IgG b) IgA c) IgD d) IgE
11. Which of the following does not participate in the formation of antigen-antibody complexes?
    a) Hydrophobic bonds b)Covalent bonds c) Electrostatic interactions d) Hydrogen bonds
12. Spontaneous activation of complement protein is observed in
    a)Classical pathway b) alternate pathway c) MBL pathway d) none of the above
13. The mechanism that characterizes the expression of MHC molecules on cell surface is:
    a) allelic exclusion b)codominant expression c)class switching d) differential RNA processing
14. The valency of IgA secreted by plasma B cell is:

a) 2 b)4 c)8 d) 10

15. In all the complement pathways C5 convertase join with C 6,7,8,9 to form

    a) membrane attack complex, b) Complement activation complex, c) Initiation complex, d) None of the above.

16. During B cell activation ITAM is required for

    a) signal transduction, b)ligand binding, c)activate tyrosine kinases, d)none of the above

17. <u>Match the following:</u> (4)

a. Hypersensitive reaction type I	1. mediated by macrophages and Tcell
b. Hypersensitive reaction type II	2. depends on FcεRI
c. Hypersensitive reaction type III	3. blood group incompatibility
d. Hypersensitive reaction type IV	4. serum sickness

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Part B **Total Marks: 50**

Section I

Q.1(a) Describe **(any two):** (4)
i) Positive selction of T cells ii) Bone marrow stromal cell for B cell maturation
iii) co-stimuatory signal for T_H cells iv) B-cell αß heterodimer

Q.1(b) Describe the process of B cell differentiation from naïve B cell to plasma and memory B cell

OR

Q.1(b) Describe the class-II MHC pathway of antigen processing and presentation. (4)

Q.1(c) Explain T cell mediated target cell killing through granzymes and perforin (2)

Q.2 (a)Explain **(any two):** (4)
i) DNA vaccine ii) multivalent vaccine iii) active and passive immunity

Q.2 (b) Explain the process degranulation of mast cell in Type-I hypersensitivity. (3)

Q.2 (c) Explain the role of complement proteins in type III hypersensitive reaction. (3)

OR

Q.2 (c) Describe type II hypersensitive reaction with one suitable example. (3)

Section II

Q.3 (a) What is primary lymphoid organs? Explain anatomy of any one in detail. (04)

OR

Q.3 (a) Explain anatomy and function of lymphnode in detail.

Q.3 (b) write a note on primary and secondary immune response. (03)

OR

Q.3 (b) What is acquired immunity ? Explain the same in brief.

Q.3 (c) Briefly describe innate immunity. (03)

Q.4 (a) Explain the exception of one gene one poly peptide hypothesis with respect to antibody diversity. (05)

OR

Q.4(a) Explain the terms allelic exclusion and class switching with respect to immunoglobulin expression.

Q.4 (b) Describe the molecular structure of immunoglobulin in brief. (03)

Q.4 (c) What is epitope ? Explain the nature of antigenic determinant for different types of immune cells. (02)

OR

Q.4 (c) What is adjuvant ? Explain it's nature and applications.

Q.5(a) What is monoclonal antibody ? Explain hybridoma technology to produce the same. **(05)**

Q.5 (b) Write in brief on immuno electrophoresis. (03)

OR

Q.5 (b) Write on working and applications of FACS.

Q.5 (c) What is ELISA ? Explain it's principle and applications. (02)